

Part A: Conceptual Foundation

Q1. What is Data Analysis?

Data Analysis is the process of collecting, cleaning, transforming, and interpreting data to discover useful information and support decision-making.

Steps of Data Analysis

1. Data Collection
2. Data Cleaning
3. Data Exploration
4. Feature Engineering
5. Model Building
6. Result Interpretation

Q2. How to Plan a Data Science Project?

Step 1

Understand the business problem.

Step 2

Collect data from different sources.

Step 3

Clean the dataset.

Step 4

Handle missing values.

Step 5

Handle outliers.

Step 6

Feature Engineering.

Step 7

Feature Scaling.

Step 8

Build ML Model.

Step 9

Evaluate Results.

Q3. How to Frame a Machine Learning Problem?

Problem Statement:

Predict whether a customer will default on a loan.

Target Variable:

default_flag

0 = No Default

1 = Default

Q4. Explain Tensor with NumPy Example

Scalar (0-D Tensor)

```
import numpy as np
scalar = np.array(10)
print(scalar)
```

```
10
```

Vector (1-D Tensor)

```
vector = np.array([10,20,30])
print(vector)
```

```
[10 20 30]
```

Matrix (2-D Tensor)

```
matrix = np.array([[1,2],[3,4]])  
print(matrix)
```

```
[[1 2]  
 [3 4]]
```

3-D Tensor

```
tensor = np.array([  
[[1,2],[3,4]],  
[[5,6],[7,8]]  
)
```

```
print(tensor)
```

```
[[[1 2]  
  [3 4]]  
  
 [[5 6]  
  [7 8]]]
```